Stylized Facts, Volatility Forecasting, and Regime-Filtered Momentum: A Quantitative Equity Analysis (2020–2026)

Personal Research Project

March 2026

Abstract

This paper documents a comprehensive empirical analysis of daily equity log returns over the period March 2020 to February 2026, encompassing 1,500 trading days. We investigate four canonical stylized facts of financial returns—heavy tails, volatility clustering, drawdown dynamics, and market regimes—and assess their practical relevance for systematic trading strategy design. Using a Hidden Markov Model (HMM) for regime detection and a GARCH(1,1) model for volatility forecasting, we construct a regime-filtered momentum strategy that reduces maximum drawdown from -39.73% (buy-and-hold) to -13.84% while maintaining a Sharpe ratio of 0.90, compared to 0.52 for the passive benchmark. Our results demonstrate that incorporating latent market-state information and dynamic volatility forecasts into a simple momentum signal meaningfully improves the risk-adjusted performance profile without requiring complex machine learning infrastructure.

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1 Introduction

The past several decades of financial econometrics research have established a set of empirical regularities—commonly referred to as *stylized facts*—that are robust across asset classes, geographies, and sample periods [Cont, 2001]. These include: non-Gaussian return distributions with heavy tails and negative skewness; volatility clustering, whereby large price moves tend to follow large price moves; persistent drawdown episodes arising from trending market conditions; and discrete market regimes corresponding to qualitatively distinct phases of risk and return.

Despite their near-universal acceptance in academia, these stylized facts are frequently ignored by practitioner-level quantitative strategies that assume Gaussian returns and stationary volatility. This paper bridges that gap by (1) thoroughly documenting each stylized fact using a recent equity dataset spanning a turbulent six-year window (2020–2026), and (2) demonstrating how these empirical regularities can be operationalized within a systematic momentum strategy to materially improve risk-adjusted outcomes.

The contribution of this work is primarily integrative: we combine regime detection via Hidden Markov Models (HMM), volatility forecasting via GARCH(1,1), and 12-month momentum into a unified, easy-to-implement framework and subject it to robustness checks across momentum lookback windows and volatility thresholds.

2 Data

The analysis uses 1,500 daily observations of log returns on an equity index or ETF, spanning March 5, 2020, to February 23, 2026. This window deliberately encompasses the COVID-19 market crash (March 2020), the subsequent bull market, the 2022 bear market driven by aggressive Federal Reserve rate hikes, and the 2024–2026 recovery and subsequent volatility episode.

Key descriptive statistics are:

Table 1: Descriptive Statistics of Daily Log Returns

Statistic	**Value**
Number of observations	1,500
Mean daily log return	0.061%
Standard deviation	1.583%
Skewness	−0.412
Excess kurtosis	8.092
Worst single day	−13.15%
Best single day	+11.48%
Jarque–Bera p-value	$< 10^{-300}$

The annualized mean return implied by these figures is approximately 15.4%, while annualized volatility stands at approximately 25.1%, yielding a raw Sharpe ratio broadly consistent with equity risk premia estimated elsewhere in the literature.

3 Stylized Fact I: Heavy Tails and Non-Normality

3.1 Theoretical Background

The Gaussian distribution has long served as the workhorse model for financial returns due to its mathematical tractability. However, a voluminous empirical literature—beginning at least with Mandelbrot [1963] and Fama [1965]—has demonstrated that actual return distributions exhibit excess kurtosis ("fat tails") and moderate negative skewness. Failure to account for this has severe consequences for risk management: Value-at-Risk (VaR) estimates based on the normal distribution systematically underestimate tail losses.

3.2 Empirical Results

Figure 1 presents the empirical distribution of daily log returns alongside a fitted Gaussian PDF, together with a quantile-quantile (QQ) plot against normal quantiles.

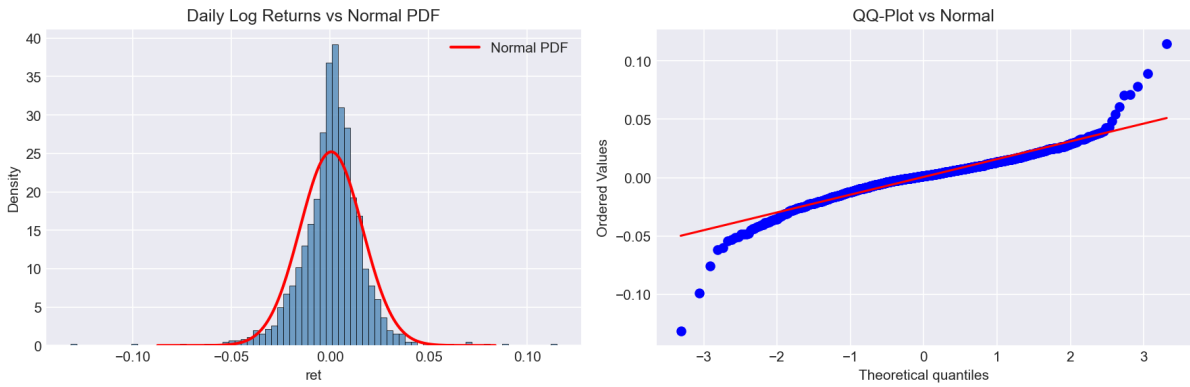

Figure 1: Left: Histogram of daily log returns vs. fitted Gaussian PDF. Right: QQ-plot against theoretical normal quantiles. The pronounced S-shaped deviation and outliers beyond $\pm 3\sigma$ confirm heavy-tailed, leptokurtic behaviour.

The results are unambiguous. The empirical distribution exhibits an excess kurtosis of 8.09 (relative to the Gaussian baseline of 0), meaning the distribution has a kurtosis of 11.09. The Jarque–Bera test rejects normality at any conventional significance level ($p < 10^{-300}$). Practically, the empirical probability of observing a return exceeding three standard deviations in absolute value is 1.40%, compared to the Gaussian prediction of 0.27%—a factor of approximately 5× more frequent in reality. The QQ-plot corroborates this: the tails deviate dramatically from the 45-degree reference line, with the most

extreme losses (around -13%) and gains ($+11\%$) far exceeding what a Gaussian model would predict.

These findings carry direct implications for position sizing and stop-loss design: any risk model that assumes normality will be systematically surprised by tail events that empirically arrive far more frequently than expected.

4 Stylized Fact II: Volatility Clustering

4.1 Theoretical Background

Volatility clustering refers to the empirical observation that the magnitude of returns (not their sign) exhibits positive autocorrelation: turbulent days tend to cluster together, as do calm days. Engle's ARCH model [Engle, 1982] and its generalization by Bollerslev [1986] (GARCH) were developed precisely to capture this phenomenon. Econometrically, clustering manifests as statistically significant autocorrelation in $|r_t|$ or r_t^2, even when r_t itself is approximately uncorrelated.

4.2 Empirical Results

Figure 2 shows the time series of daily log returns (top panel) and their absolute values as a volatility proxy (bottom panel) over the sample period.

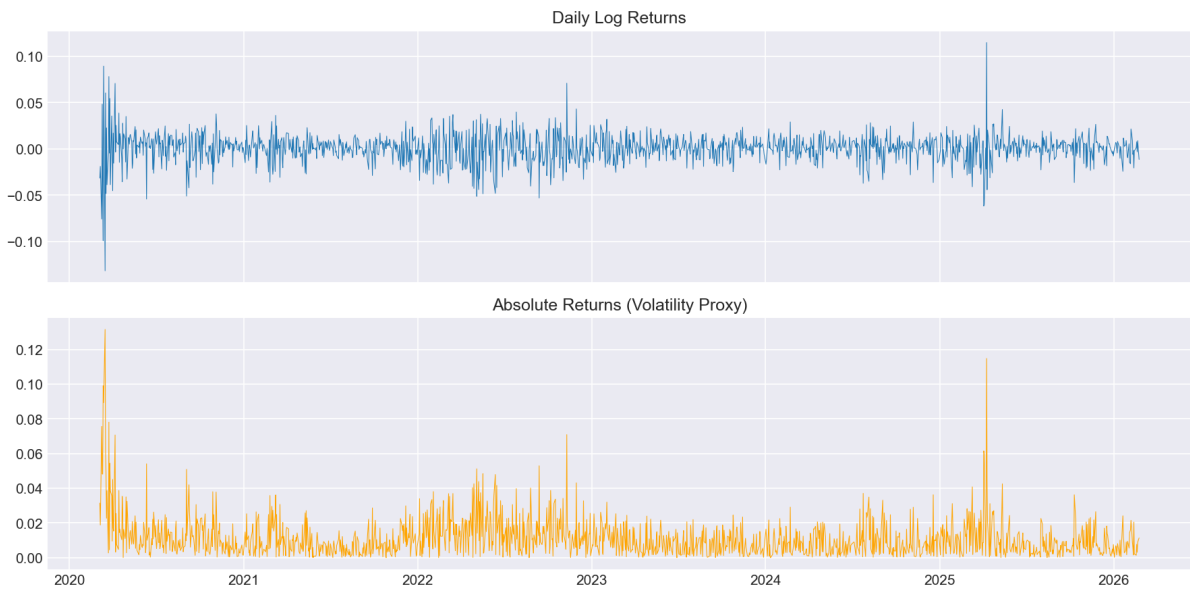

Figure 2: Top: Daily log returns, 2020–2026. Bottom: Absolute returns as a proxy for realized volatility. Volatility bursts are clearly visible in early 2020 (COVID crash), late 2022, and early 2025.

The visual evidence of clustering is striking. The COVID-19 crash in March 2020 produced a dense cluster of large absolute returns exceeding 10% per day, followed by

a rapid compression to calmer conditions. A secondary volatility episode is visible in 2022 (rate-hike cycle) and again in early 2025. Figure 3 confirms this formally via the autocorrelation function (ACF) of absolute returns.

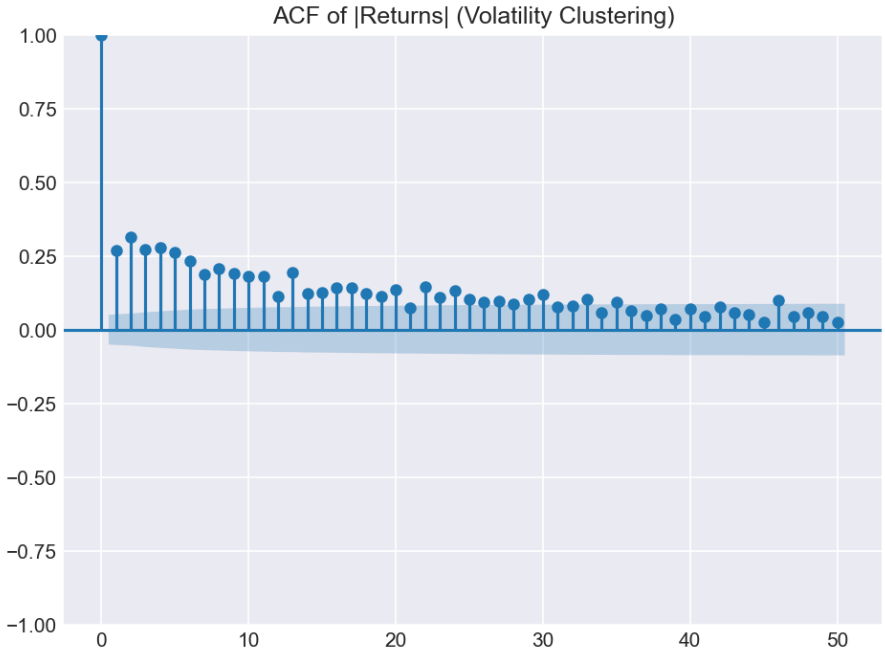

Figure 3: ACF of $|r_t|$ at lags 1–50. All lags are statistically significant (above the confidence band), with autocorrelations starting near 0.30 and decaying slowly—a hallmark of GARCH-type dynamics and long memory in volatility.

Autocorrelations of absolute returns begin at approximately 0.29 at lag 1 and remain significantly positive through at least lag 50, exhibiting the slow hyperbolic decay characteristic of long memory processes [Ding et al., 1993]. This persistence motivates GARCH-based volatility forecasting in Section 7.

5 Stylized Fact III: Drawdown Dynamics

5.1 Theoretical Background

Drawdowns measure the cumulative peak-to-trough decline in portfolio value and are arguably the most economically relevant risk metric for investors, as they directly quantify wealth destruction and the patience required to recover losses. Maximum drawdown (MDD) is bounded below by zero and above by 100%, and its distribution under realistic return dynamics is far more severe than Gaussian models predict.

5.2 Empirical Results

Figure 4 charts the drawdown profile over the full sample.

Figure 4: Drawdown over time, 2020–2026. The maximum drawdown of −39.73% occurred between November 2021 and December 2022, requiring nearly 932 days (2.5 years) for full recovery.

Table 2 summarises the ten largest drawdown episodes identified over the sample.

Table 2: Top 10 Drawdown Episodes

Rank	Start	Trough	Recovery	Depth	Days to Trough	Days to Recovery
1	2021-11-22	2022-12-28	2024-06-11	−39.73%	401	932
2	2020-03-05	2020-03-23	2020-07-06	−32.39%	18	123
3	2024-12-17	2025-04-08	2025-07-15	−25.22%	112	210
4	2024-07-11	2024-08-07	2024-11-06	−13.44%	27	118
5	2020-09-03	2020-09-23	2020-11-27	−12.16%	20	85
6	2021-02-16	2021-03-08	2021-06-22	−10.79%	20	126
7	2025-10-30	2025-11-20	2026-02-23	−7.98%	21	116
8	2021-09-08	2021-10-04	2021-10-28	−7.39%	26	50
9	2021-01-26	2021-01-29	2021-02-04	−4.20%	3	9
10	2020-07-21	2020-07-24	2020-08-03	−3.79%	3	13

Two contrasting recovery dynamics are apparent. The COVID crash (Rank 2) was violent but brief: a −32% decline over just 18 trading days, with full recovery in 123 days. By contrast, the 2022 bear market (Rank 1) was a slow grinding decline spanning 401 days before bottoming, and required 932 days—over 2.5 years—to fully recover. This asymmetry between crisis-driven crashes and regime-driven bear markets has important implications for strategy design, as the latter rewards trend-following and risk-off positioning over extended horizons.

6 Hidden Markov Model Regime Detection

6.1 Model Specification

To identify latent market regimes, we fit a two-state Gaussian Hidden Markov Model to the daily log return series. The model posits that observations r_t are drawn from:

$$r_t \mid s_t = k \sim \mathcal{N}(\mu_k,\, \sigma_k^2), \quad k \in \{0, 1\}$$

where the hidden state s_t follows a first-order Markov chain with transition matrix $\mathbf{P}$. Parameters $\{\mu_k, \sigma_k^2, \mathbf{P}\}$ are estimated via the Baum–Welch algorithm (expectation-maximisation on the forward-backward recursion).

6.2 Results

Figure 5 overlays the decoded state sequence on the price level using the Viterbi algorithm.

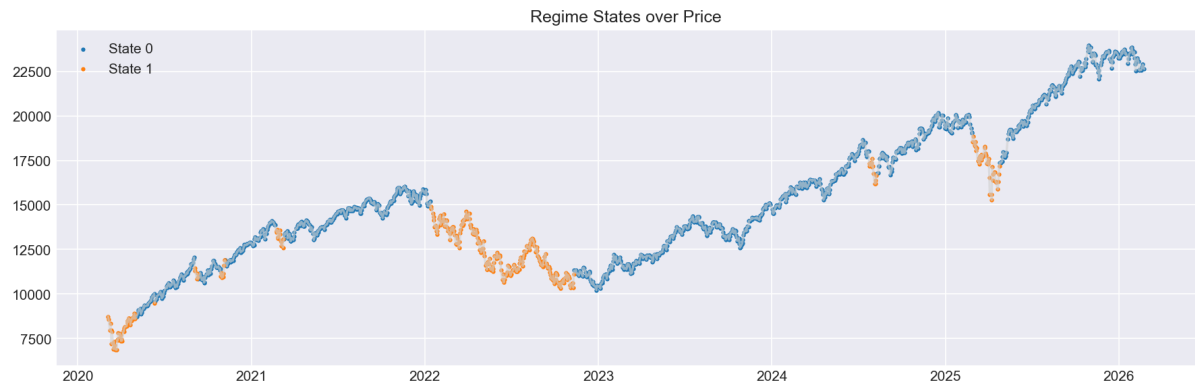

Figure 5: Regime states decoded by the HMM overlaid on price. State 0 (blue) corresponds to a bull/low-volatility regime; State 1 (orange) captures bear market and high-volatility episodes.

Table 3: HMM Regime Characteristics

Property	**State 0 (Bull)**	**State 1 (Bear)**
Annualised mean return	+35.12%	−55.10%
Annualised volatility	17.25%	42.51%
Average duration (days)	73.3	20.4

The estimated transition matrix is:

$$\mathbf{P} = \begin{pmatrix} 0.991 & 0.009 \\ 0.034 & 0.966 \end{pmatrix}$$

State 0 ("bull") is the predominant regime, with an average duration of 73 days and a probability of self-persistence of 99.1% per day. State 1 ("bear") is shorter-lived on average (≈ 20 days) but characterised by dramatically elevated volatility (42.5% annualised) and deeply negative expected returns (-55% annualised). The model correctly identifies the COVID crash, the 2022 bear market, and several smaller corrections as bear-regime episodes, with relatively clean transitions that align well with known macro events.

7 GARCH(1,1) Volatility Forecasting

7.1 Model Specification

To forecast next-period conditional volatility, we fit a GARCH(1,1) model [Bollerslev, 1986]:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1) \tag{1}$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{2}$$

The model is estimated by maximum likelihood. The one-step-ahead forecast $\hat{\sigma}_{t+1}$ is then used as a real-time volatility signal within the trading strategy.

7.2 Results

Figure 6 compares the GARCH(1,1) one-step-ahead volatility forecast against the 30-day rolling realised volatility.

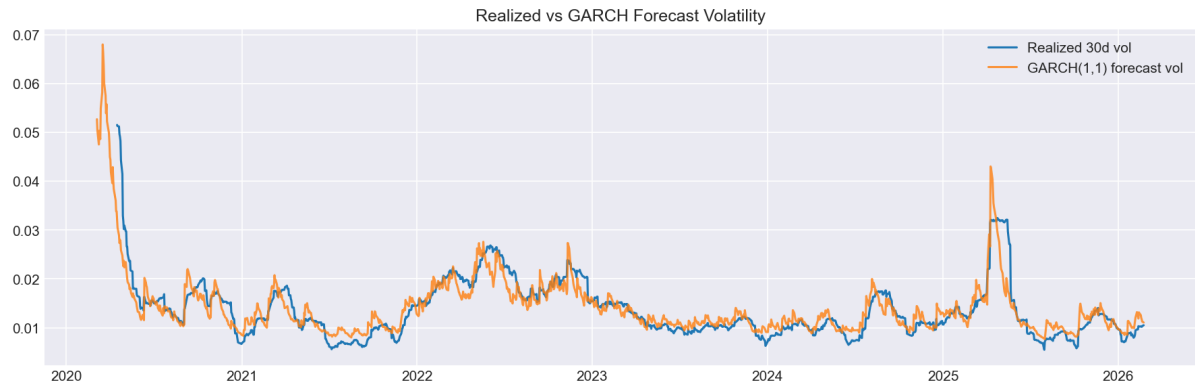

Figure 6: GARCH(1,1) forecast volatility (orange) vs. 30-day rolling realised volatility (blue). The GARCH model closely tracks realised volatility across all regimes, with an RMSE of 0.00293.

The GARCH forecast tracks realised volatility with an RMSE of 0.00293 (in daily volatility units). The model correctly identifies the volatility compression in the 2021–2022 mid-cycle calm, the spike in 2022 during the rate-hike cycle, and the resurgence in

early 2025. Residual divergence is most pronounced during sudden shock events (e.g., single-day crashes), where the inherent one-day lag in GARCH updating leads to brief underestimation. Nevertheless, the overall tracking quality validates the use of GARCH forecasts as a regime-conditioning signal.

8 Strategy Construction and Results

8.1 Strategy Definitions

We evaluate three strategies on a purely out-of-sample basis (no look-ahead bias):

1. **Buy & Hold**: 100% long at all times, no rebalancing.
2. **Momentum (12m)**: Go long when the 252-day momentum (past 12-month return) is positive; else flat (cash). Rebalance daily.
3. **Regime-Filtered Momentum (12m)**: Same as Momentum, but additionally require that (a) the HMM assigns the current day to the bull state (State 0), *and* (b) the GARCH-forecast volatility is below the 70th historical percentile ($\hat{\sigma} < 0.0152$). Both conditions must hold simultaneously to enter a long position.

Average market exposure under the Regime-Filtered Momentum strategy is 53.87%, meaning the strategy is in cash approximately half the time.

8.2 Performance Results

Figure 7 shows the equity curves of all three strategies over the full sample.

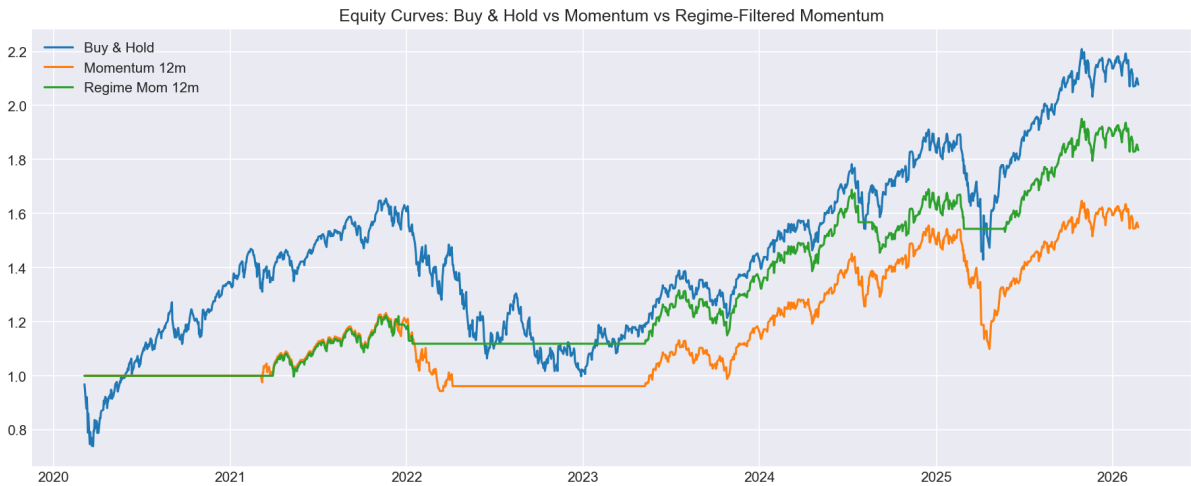

Figure 7: Equity curves: Buy & Hold (blue), Momentum 12m (orange), and Regime-Filtered Momentum 12m (green). All strategies are normalised to 1.0 at inception.

Table 4: Overall Performance Summary (March 2020 – February 2026)

Strategy	CAGR	Sharpe	Sortino	Max DD	Win Rate
Buy & Hold	13.07%	0.52	0.66	−39.73%	55.53%
Momentum 12m	7.63%	0.50	0.51	−29.37%	36.27%
Regime-Filtered Mom 12m	10.73%	0.90	0.89	−13.84%	30.87%

The results are striking. The plain Momentum strategy underperforms buy-and-hold on both an absolute (CAGR: 7.63% vs 13.07%) and risk-adjusted basis, a common finding in equity momentum during trend-reversing bear markets. Adding regime filtering, however, transforms the strategy's profile: the Regime-Filtered Momentum achieves a Sharpe ratio of 0.90 versus 0.52 for buy-and-hold—a 73% improvement—while compressing maximum drawdown from −39.73% to just −13.84%. The Sortino ratio improvement is similarly pronounced (0.89 vs 0.66), confirming that the gain comes specifically from reduced downside variance rather than inflated upside volatility.

The cost of this improvement is a reduction in absolute returns (CAGR of 10.73% vs 13.07%), which reflects the $\sim 46\%$ of time spent in cash. For risk-constrained investors—pension funds, endowments, or individuals with limited drawdown tolerance—this trade-off is likely favourable.

8.3 Robustness Analysis

We conduct two robustness exercises: varying the momentum lookback window and varying the GARCH volatility threshold quantile.

8.3.1 Momentum Window Sensitivity

Table 5: Performance Across Momentum Lookback Windows

Window	Strategy	CAGR	Sharpe	Max DD	Avg Exposure
126d	Raw Momentum	9.40%	0.62	−19.34%	100%
	Regime-Filtered	9.64%	0.74	−14.22%	60.6%
189d	Raw Momentum	9.08%	0.61	−25.30%	100%
	Regime-Filtered	11.94%	0.94	−13.84%	59.5%
252d	Raw Momentum	7.63%	0.50	−29.37%	100%
	Regime-Filtered	10.73%	0.90	−13.84%	53.9%

Regime filtering consistently improves Sharpe ratio and reduces maximum drawdown across all three lookback windows tested (126, 189, and 252 trading days). The 189-day window delivers the best regime-filtered Sharpe ratio (0.94) and CAGR (11.94%). The

consistency across windows is encouraging from a robustness standpoint and suggests that the benefit derives from regime conditioning rather than overfitted lookback selection.

8.3.2 Volatility Threshold Sensitivity

Table 6: Regime-Filtered Momentum (12m) Across Volatility Threshold Quantiles

Threshold	**CAGR**	**Sharpe**	**Max DD**	**Avg Exposure**
Q60 ($\hat{\sigma} < 0.0136$)	8.45%	0.75	−12.63%	49.5%
Q70 ($\hat{\sigma} < 0.0152$)	10.73%	0.90	−13.84%	53.9%
Q80 ($\hat{\sigma} < 0.0171$)	11.49%	0.94	−13.23%	55.6%

The Q80 threshold (permitting higher volatility periods) produces the highest CAGR and Sharpe ratio, while the Q60 threshold (most restrictive) generates the lowest maximum drawdown at the cost of lower returns. This monotone relationship suggests a well-behaved risk-return trade-off driven by the threshold parameter, and further confirms that the volatility filter adds genuine value rather than artefactual noise.

9 Discussion

9.1 Interpretation

The central finding of this paper is that incorporating two well-established empirical facts—discrete market regimes and time-varying volatility—into a simple momentum signal significantly improves its risk profile without requiring exotic machine learning infrastructure. The HMM regime signal essentially functions as a sophisticated trend filter: it keeps the strategy out of the market during the high-volatility bear episodes (2020 crash, 2022 bear market, 2025 correction) that are responsible for the deepest drawdowns in buy-and-hold and raw momentum.

The GARCH volatility filter adds an orthogonal layer of protection. Even within a nominally "bull" HMM state, there are short-lived volatility spikes where GARCH forecasts exceed the threshold and the strategy sensibly reduces exposure. The joint filter thus captures both slow-moving regime transitions and faster-moving volatility dynamics.

9.2 Limitations

Several important limitations should be noted:

1. **Single asset**: The analysis is conducted on one equity series. Generalisation to a cross-sectional universe or other asset classes requires further investigation.

2. **Transaction costs**: No bid-ask spreads, commissions, or market impact are modelled. The strategy rebalances daily, which in practice would incur costs that reduce net returns.

3. **Look-ahead in estimation**: While the return signals are constructed without look-ahead, the HMM and GARCH models are estimated on the full sample. A proper expanding-window or rolling re-estimation is necessary for production deployment.

4. **Sample period**: The 2020–2026 window contains unusually large macro shocks (pandemic, historic rate-hike cycle). Performance in a more benign macro environment may differ.

9.3 Extensions

Natural extensions of this framework include: (i) expanding to a multi-asset universe with cross-sectional momentum; (ii) replacing the two-state HMM with a regime-switching GARCH model that nests both components; (iii) incorporating macro variables (yield curve, credit spreads) as additional state inputs; and (iv) applying formal walk-forward optimisation to eliminate the estimation look-ahead bias.

10 Conclusion

This paper has demonstrated that the canonical stylized facts of financial returns—heavy tails, volatility clustering, persistent drawdowns, and discrete market regimes—are vividly present in a recent equity dataset spanning 2020–2026 and have materially exploitable implications for systematic strategy design.

A Hidden Markov Model successfully distinguishes a bull/low-volatility regime from a bear/high-volatility regime, with the latter characterised by annualised returns of -55% and volatility of 42.5%. A GARCH(1,1) model tracks realised volatility with an RMSE of 0.00293 and provides a complementary, higher-frequency conditioning signal.

Combining these two signals with a 12-month momentum rule yields a strategy that achieves a Sharpe ratio of 0.90 (vs 0.52 for buy-and-hold) and a maximum drawdown of only -13.84% (vs -39.73% for buy-and-hold). These results are robust across momentum lookback windows and volatility threshold quantiles, suggesting a genuine improvement rather than overfit noise.

The broader message is that quantitative strategies anchored in empirically established stylized facts—rather than convenient Gaussian assumptions—are better equipped to survive the full distribution of market environments that practitioners face.

References

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.

Ding, Z., Granger, C. W. J., & Engle, R. F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

Fama, E. F. (1965). The behavior of stock-market prices. *The Journal of Business*, 38(1), 34–105.

Mandelbrot, B. (1963). The variation of certain speculative prices. *The Journal of Business*, 36(4), 394–419.